Signal Support in Joint Operations

Signal Officer Advance Course

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Instructor Profile



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Educational Attainment:

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References

a. Philippine Army Manual (PAM) 6-00 Command and Control Communications, and Cyber Systems (2nd Edition) dated 11 December 2017;

b. PAM 6-0102 Signal Operations Manual dated 15 April 2022;

c. PAM 6-0102-04 Cyberspace Operations dated 24 February 2020;

d. PAM 3-00 Army Operations dated 06 May 2014; and

e. PAM 3-03 Combined Arms Operations dated 17 July 2017.

TERMINAL LEARNING OBJECTIVES

At the end of the subject, the students will be able to determine signal support in joint operations IAW PAM 6-0102. (C3)

ENABLING LEARNING OBJECTIVES

- Discuss the Role and Importance of Signal Support in Joint Operations. (C3)
- Determine Signal Support Systems and Technologies. (C3)
- Discuss Principles of Signal Support in Joint Operations. (C3)
- Discuss Tactical Considerations in Signal Support Operations. (C3)
- Determine the Plan and Coordinate Signal Support in Joint Operations. (C3)
- Determine the security measures and protocols in Signal Support Operations. (C3)

Joint Operations

military operations involving two or more AFP services working under a single commander to accomplish a common mission.

Service	Contribution to a joint operation
Philippine Army	Provides land forces, including combined arms teams, to conduct land operations and support air or maritime operations.
Philippine Navy	Provides maritime capabilities, including forces for sea control.
Philippine Air Force	Provides air capabilities, including forces for counter-air operations.

Signal Support connects the Joint Force

Army, Navy, and Air Force elements must be able to communicate and exchange the information needed to coordinate their actions.

Remember this distinction: infantry, armor, artillery, and engineers working together illustrate combined arms operations. Army forces working with Navy or Air Force forces under a common commander illustrate joint operations. An Army combined arms team can therefore be part of a larger AFP joint operation.

Operational Environment



ROLES OF SIGNAL SUPPORT IN MILITARY OPERATIONS

Philippine Setting



Roles of Signal Support to Military Operations

Signal support enables the participating AFP services to communicate, coordinate their actions, and maintain effective command and control during joint operations. When the Philippine Army operates with the Philippine Navy or Philippine Air Force, their communication systems and procedures must allow them to work together.

1. Connect the participating Services
2. Enable Effective Command and Control
3. Facilitate coordination between operating troops and supporting assets
4. Support the synchronization of joint activities
5. Apply communication protection measures

Roles of Signal Support to Military Operations

1. **Connect the participating services;**

   The important term is interoperability. It means that the participating services can communicate effectively despite differences in their communication equipment or procedures.

Example: Army troops conducting an island operation need to communicate with a supporting Navy element. Providing compatible communication equipment helps both elements coordinate their activities.

Why it matters: A joint plan depends on the services being able to exchange information and instructions.

Roles of Signal Support to Military Operations

2. Enable effective command and control

Signal support allows commanders to communicate decisions and coordinate the capabilities assigned to the operation.

Example: An Air Force radio operator working at an Army tactical operations center helps the commander communicate with supporting air assets.

What you should remember: The commander needs timely communication to direct forces and adjust their actions as the situation changes

Roles of Signal Support to Military Operations

3. Facilitate coordination between operating troops and supporting assets

Army radio operators may be embedded with another service. These operators act as a relay between operating units and the assets provided by that service.

Example: An Army radio operator assigned alongside a naval element relays information between Army troops ashore and the naval support element.

What you should remember: This arrangement helps bridge communication gaps between the forces conducting the task and the forces supporting them.

Roles of Signal Support to Military Operations

4. Support the synchronization of joint activities

The Army forces integrate and synchronize with other services through joint planning. It also describes how services receive supported or supporting roles according to the operation.

Example: In an island operation, the Navy transports troops, the Air Force provides assigned air support, and the Army conducts land tasks. Communication enables them to exchange readiness reports and coordinate changes.

What you should remember: Each service's contribution must occur at the appropriate time and support the common mission.

Roles of Signal Support to Military Operations

5. Apply communication protection measures

It operates in a less secure mode and is susceptible to attack and eavesdropping. It identifies ciphertext and double talk as measures to mitigate this vulnerability.

What you should remember: Establishing communication must be accompanied by attention to its security. The measures as mitigation; it does not describe the vigilance net as fully secure.

Roles of Signal Support to Military Operations

Role of signal support	Importance to the joint force
Connect the services	Enables communication interoperability.
Support command and control	Links commanders with operating units and supporting assets.
Relay information across services	Bridges communication gaps.

The key point I want you to understand is this: signal support turns the joint force's coordinated plan into coordinated action by keeping its commanders, operating troops, and supporting assets connected.

SIGNAL SUPPORT SYSTEMS AND TECHNOLOGIES

Philippine Military Setting

1. Tactical Radio Systems



2. Satellite Communications



3. Line and Fiber Communications



4. Data Networks





5. Radio Relay Systems



6. Command and Control Systems



7. Communications Security



8. Power and Support Equipment



Signal Support Systems and Technologies

Signal support systems and technologies are the communication networks, equipment, and information services that enable commanders and units to exchange information and exercise command and control.

1. Fixed Communications System
2. Tactical Communications System
3. Special Purpose Communications Systems
4. Operations-center systems
5. Network support and cyber protection

Signal Support Systems and Technologies

1. Fixed Communications System

The communication backbone. Fixed communication combines wired and wireless technologies to support the Army's voice and data communications.

Example: An Army headquarters uses its communication services to exchange reports and participate in coordination meetings with the joint headquarters. Connection to another service still requires approved arrangements and compatible systems.

Signal Support Systems and Technologies

2. Tactical communication systems

Communication for operating troops. Tactical communication as military-grade communication used both on the move and on the halt..

Example: Army troops ashore need communication with a supporting naval element. Signal personnel must establish a compatible link or an operator-relay arrangement.

Signal Support Systems and Technologies

3. Special purpose communication systems

Commercial devices used to augment fixed and tactical communications, particularly for routine or administrative communication

Example: A commercial video-conferencing platform may support an authorized administrative meeting. The information discussed must remain within what the applicable policies permit.

Signal Support Systems and Technologies

4. Operations-center systems — bring information to the commander

An operations center as the facility where information from sensors and communication devices becomes available to the commander.

Example: A tactical operations center receives Army reports and coordinates with supporting air and naval elements through established communication links.

Signal Support Systems and Technologies

5. Network support and cyber protection

The following activities that sustain and protect the information networks supporting operations. Information Network Operations (INO), Defensive Cyberspace Operations (DCO).

These are supporting activities, rather than additional categories of communication equipment. They help keep the network services available and protected.

How do we determine which systems to use in joint operations?

We begin with the requirement, then select the technology. Require a signal estimate, signal planning, and electromagnetic spectrum management. Emphasizes communication interoperability among the major services

1. Who needs to communicate? Identify the headquarters, operating troops, and supporting service elements.
2. What must they exchange? Determine whether they need voice, reports, data, or other information.
3. Where will they operate? Consider movement and the environment's effects on equipment.
4. Can their systems work together? Check equipment compatibility and agreed procedures.
5. How will communication be protected, sustained, and restored? Include security and recovery arrangements.

How do we determine which systems to use in joint operations?

Where direct interoperability is difficult, identify practical arrangements: lend communication equipment, embed Army radio operators with another service, or embed another service's operators at the Army tactical operations center.

Remember: Fixed systems provide the backbone, tactical systems connect operating troops, and special purpose systems provide additional options. Their effectiveness in joint operations depends on compatible equipment, trained operators, coordinated procedures, and appropriate protection.

PRINCIPLES OF SIGNAL SUPPORT

IN COMBINED ARMS OPERATIONS • PHILIPPINE SETTING

1. INTEGRATION

Link all combined arms elements



2. INTEROPERABILITY

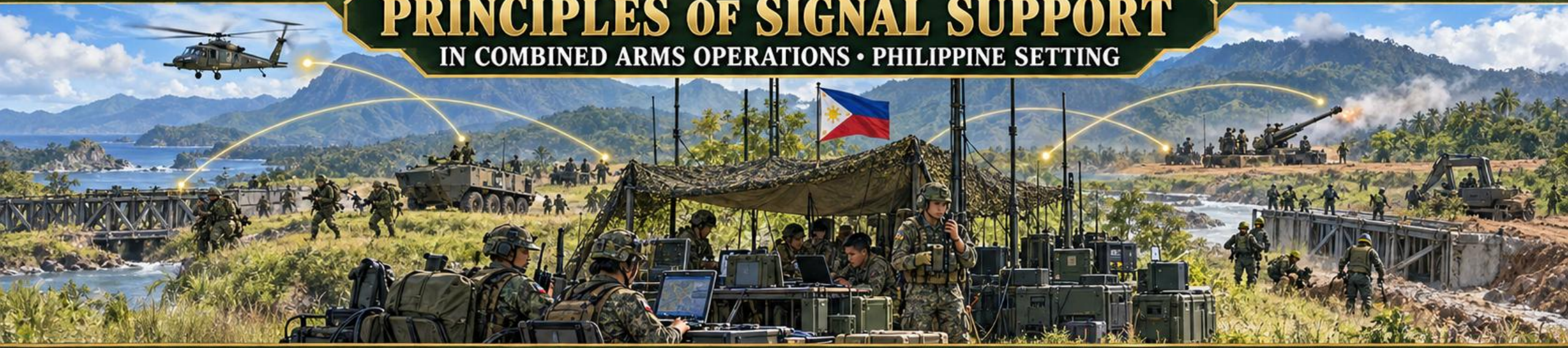
Enable systems and units to communicate



3. RELIABILITY AND CONTINUITY

Maintain dependable communications





4. FLEXIBILITY AND MOBILITY

Adapt to changing missions and movement



5. SECURITY

Protect communications and information



6. REDUNDANCY AND RECOVERY

Provide alternatives and restore service



Principles of Signal Support in Joint Operations

These principles guide signal support across its services, including support to joint operations. In joint operations, we apply them to keep the Army, Navy, and Air Force connected so they can accomplish a common mission.

Principles of Signal Operations

1. Operations Focused
2. Interoperable
3. Redundant
4. Scalable
5. Secured

Principles of Signal Support in Joint Operations

Principles of Signal Operations

1. Operations Focused

Signal support should never become the factor that limits the commander's ability to execute the mission.

Example: Army troops need to remain connected with a naval transport element during an island operation. Signal support must provide the equipment, power, and operator support needed throughout that task.

Remember: Start with the operational requirement when choosing communication equipment.

Principles of Signal Support in Joint Operations

Principles of Signal Operations

2. Interoperable

Army units must be able to communicate within the Army and with other major services.

Example: An Army tactical operations center must communicate with supporting air assets. Where direct communication is difficult, the lending equipment or embedding radio operators as possible arrangements.

Remember: Owning radios is insufficient; the participating services must establish a working communication link.

Principles of Signal Support in Joint Operations

Principles of Signal Operations

3. Redundant

Redundancy supports continued communication through multiple transmission paths, backups, self-healing strategies, and data replication.

Example: A joint headquarters has an approved alternative communication link available if its main connection fails. Operators know how to change to that alternative.

Remember: A backup must be available and usable when the main means fails.

Principles of Signal Support in Joint Operations

Principles of Signal Operations

4. Scalable

Signal support to expand its services according to the situation.

Example: An operation initially involves Army and Navy elements. When Air Force elements join, signal personnel adjust the communication arrangements to include them.

Remember: Communication support must adapt as the force and mission change.

Principles of Signal Support in Joint Operations

Principles of Signal Operations

5. Secured

The security remains a guiding principle regardless of the communication technology or service used.

Connections between services must follow applicable security requirements. A convenient or interoperable system still needs appropriate protection.

Remember: Assess both the ability to communicate and the protection of the information being exchanged.

Principles of Signal Support in Joint Operations

Principle	Question to ask
Operations focused	Does communication support the commander's mission requirements?
Interoperable	Can the participating services communicate effectively?
Redundant	Can essential communication continue if a link fails?
Scalable	Can support adapt when forces or requirements change?
Secured	Does communication follow applicable security policies?

Remember: effective joint signal support must serve the mission, connect the services, survive failures, adapt to change, and protect information. Plan a system that can grow and adapt with the joint force.

TACTICAL CONSIDERATIONS

IN SIGNAL SUPPORT OPERATIONS • PHILIPPINE SETTING

MISSION REQUIREMENTS
Match communications to the mission



TERRAIN AND WEATHER
Account for obstacles and conditions



THREATS AND PROTECTION
Protect personnel, equipment and information



MOBILITY AND SITE SELECTION
Support movement and suitable positioning



COORDINATION AND INTEROPERABILITY
Connect supported units and systems



SPECTRUM AND INTERFERENCE
Manage authorized use and signal interference



LOGISTICS AND POWER
Sustain equipment, energy and maintenance



CONTINUITY AND RECOVERY
Prepare alternatives and restore communications



Tactical Considerations in Signal Support Operations

Tactical considerations in signal support operations are the practical factors that commanders and signal personnel must assess to keep communications effective during a mission.

They include the mission, terrain, movement, equipment, personnel, threats, security, and recovery arrangements.

Tactical Considerations in Signal Support Operations

1. Mission and commander's intent
2. Terrain, distance, and weather
3. Unit movement and communication-site location
4. Interoperability in joint operations
5. Equipment, personnel, and sustainment
6. Spectrum management and electronic threats
7. Communication and cyber security
8. Redundancy and recovery
9. Priorities when resources are limited

Tactical Considerations in Signal Support Operations

1. Mission and commander's intent

Determine what the force must accomplish and what communication it needs to accomplish it. Signal planners to obtain requirements from the commander, staff, higher headquarters, subordinate and adjacent units, and applicable operating instructions. Communication should support mission execution without becoming its limiting factor.

Example: If Army troops must coordinate with naval transport and air support, the signal plan must include those links alongside communication within the Army.

Tactical Considerations in Signal Support Operations

2. Consider terrain, weather, and signal coverage

- Several challenges in the Philippine environment:
- Separation between islands makes connecting dispersed units difficult.
- Mountains affect line-of-sight communication.
- Dense vegetation affects signal propagation.
- Urban structures and infrastructure influence radio communication.
- Storms and typhoons can render communication assets inoperable.

Ask: Will the selected communication means work in the actual operating area?

Tactical Considerations in Signal Support Operations

3. Unit movement and communication-site location

Communication support must follow the force's changing tactical disposition. Communication both on the move and on the halt

Example: When a command post relocates, its communication support must be arranged so that the commander can continue directing the operation.

Ask: Can communication continue as troops and headquarters move?

Tactical Considerations in Signal Support Operations

4. Interoperability in joint operations

The Army, Navy, and Air Force must be able to exchange information through compatible systems and agreed procedures.

Example: If Army and naval equipment cannot communicate directly, an approved equipment-sharing or operator-relay arrangement may provide the necessary connection.

Ask: Have the participating services established a working communication link?

Tactical Considerations in Signal Support Operations

5. Equipment, personnel, and sustainment

A plan must reflect the resources actually available. Assessment of equipment serviceability, maintenance readiness, personnel availability and training, and funding for installation, operation, and maintenance.

Example: A unit has enough radios, but insufficient charged batteries or trained operators to sustain their use throughout the mission.

Ask: Can the available equipment and personnel maintain communication for the required duration?

Tactical Considerations in Signal Support Operations

6. Spectrum management and electronic threats

Planning, coordinating, and managing the electromagnetic spectrum because units use different technologies and frequencies.

The tactical communication is susceptible to jamming and interception, requiring protection measures in the signal plan.

Ask: Is spectrum use coordinated, and does the plan address interference and electronic threats?

Tactical Considerations in Signal Support Operations

7. Communication and cyber security

Planning for cyberattacks and compromised communication frequencies, including network monitoring, incident-response arrangements, and procedures for changing compromised frequencies.

Example: Lost communication equipment creates a security concern as well as an equipment shortage. The unit must follow the prescribed reporting and protective procedures.

Ask: How will the force protect its information and respond to a compromise?

Tactical Considerations in Signal Support Operations

8. Redundancy and recovery

Redundant communication, equipment repair and replacement, and technical support when connectivity is interrupted.

Example: If the primary link fails, operators use the planned alternative while technical personnel diagnose and restore the affected system.

Ask: What happens when the main communication means fails?

Tactical Considerations in Signal Support Operations

9. Priorities when resources are limited

Planners to compare requirements with capabilities. When resources are insufficient, support priorities must reflect their importance to mission accomplishment.

Ask: Which communication requirements are most critical to the commander’s mission?

Tactical Considerations in Signal Support Operations

The Connect–Sustain–Recover (CSR) framework ties these considerations together:

Framework	Tactical application
Connect	Establish the required links between commanders, troops, headquarters, and supporting services.
Sustain	Maintain those links despite movement, environmental conditions, and resource demands.
Recover	Restore communication after failure, damage, or attack.

As a signal planner is to ensure that the force can establish communication, maintain it throughout the mission, and restore it when disrupted.

PLAN AND COORDINATE SIGNAL SUPPORT

IN COMBINED ARMS OPERATIONS • PHILIPPINE SETTING

1. ANALYZE THE MISSION



Identify tasks and information needs

2. ASSESS SIGNAL REQUIREMENTS



Determine personnel, systems and coverage needs

3. DEVELOP THE SUPPORT PLAN



Plan connectivity, protection and alternatives



4. COORDINATE WITH ALL ELEMENTS



Align support with maneuver, fires and sustainment

5. REHEARSE AND VERIFY



Check links, procedures and readiness

6. MONITOR AND ADJUST



Maintain support as the operation changes

Plan and Coordinate Signal Support in CAO

Planning and coordinating signal support in joint operations means determining the communication requirements of the joint force, organizing the available resources, and ensuring that the participating services can communicate throughout the mission

Plan and Coordinate Signal Support in CAO

1. Prepare the signal estimate: understand the situation
2. Determine communication requirements
3. Determine available resources
4. Establish support priorities
5. Develop recovery procedures
6. Prepare and coordinate the tentative plan
7. Validate and finalize the plan
8. Publish the plan and maintain coordination

Plan and Coordinate Signal Support in CAO

1. **Prepare the Signal Estimate: Understand the situation**

   Determine the situation before selecting communication systems. Use METAL to guide the Signal Estimate.

Example: For an island operation involving Army, Navy, and Air Force elements, assess the required service-to-service links, separation between forces, equipment compatibility, available operators, and sustainment needs.

Plan and Coordinate Signal Support in CAO

2. Determine Communication Requirements

The planner to obtain guidance from:

- ❑ The commander.
- ❑ Other staff members.
- ❑ Higher headquarters, subordinate units, and adjacent units.
- ❑ SOPs.
- ❑ CEOI and CESI, which provide operating instructions and explain their use.

Lesson: Identify who must communicate with whom, what information they must exchange, and when the connection is required.

Plan and Coordinate Signal Support in CAO

3. Determine available resources

Resource	What to check
Equipment	Availability, serviceability, maintenance readiness, reserves, and capabilities.
Personnel	Availability, training, shortages, existing commitments, and reserves.
Funds	Support for installation, operation, maintenance, and related requirements.

Plan around resources that can actually be provided and sustained.

Plan and Coordinate Signal Support in CAO

4. Establish priorities

Comparing requirements with available capabilities. If resources are insufficient, prioritize communication support according to its importance to mission accomplishment

Example: The commander may prioritize essential command links and coordination with supporting assets before less urgent administrative requirements. The priorities must follow the mission and commander's guidance.

Plan and Coordinate Signal Support in Joint Operations

5. Develop recovery procedures

- ❑ Interrupted connectivity: redundant communication, repair, replacement, and technical support.
- ❑ Electronic warfare: protection against jamming and interception.
- ❑ Cyberattacks: network monitoring and incident-response procedures.
- ❑ Compromised frequencies: prescribed procedures for changing affected frequencies.

Lesson: Determine how communication will continue and be restored before a disruption occurs.

Plan and Coordinate Signal Support in CAO

6. Prepare the tentative signal plan

For joint operations, coordination must resolve how the services will communicate. Emphasize interoperability and identify equipment lending and embedded radio operators as possible arrangements. Spectrum use must also be coordinated.

Plan and Coordinate Signal Support in CAO

6. Prepare the tentative signal plan

Coordination requirement	What must be agreed
Command relationships	Which headquarters directs the operation and which components support each task.
Communication links	Which units and headquarters require connections.
Interoperability	Compatible equipment, configurations, procedures, and operator arrangements.
Spectrum use	Coordinated frequency requirements under applicable instructions.
Responsibilities	Who installs, operates, maintains, and restores each link.
Security and recovery	Applicable protection measures and actions during disruption.

Plan and Coordinate Signal Support in CAO

7. Validate and finalize the plan

Checking that the plan:

- Meets the command's requirements.
- Is practical and adequately resourced.
- Reflects established priorities.
- Defines responsibilities.
- Uses surveyed or reconnoitered locations.
- Identifies communication coverage and gaps.
- Includes security measures.

Verify the proposed connections and resolve identified gaps with the participating services before relying on them during execution.

Plan and Coordinate Signal Support in CAO

8. Publish the plan and maintain coordination

Publishing signal plans and orders with the necessary diagrams and instructions, including the C4S annex to the operations order or plan, as required.

During execution, continue assessing communication requirements and updating the estimate as conditions change. Apply the manual's Connect–Sustain–Recover framework: establish the links, maintain them throughout the operation, and restore them after disruption.

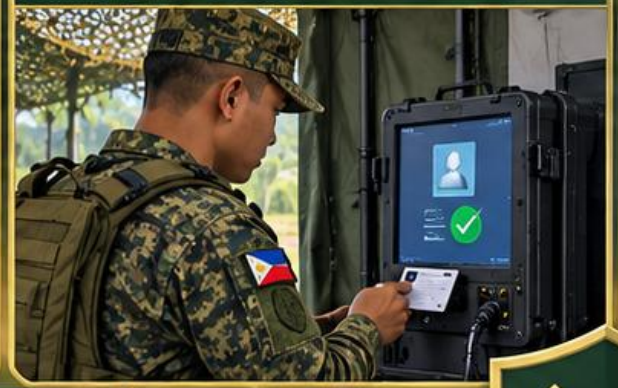
SECURITY MEASURES AND PROTOCOLS

IN SIGNAL SUPPORT OPERATIONS • PHILIPPINE SETTING

1. COMMUNICATIONS SECURITY
Use approved secure communications



2. AUTHENTICATION AND ACCESS
Verify identities and limit access



3. INFORMATION PROTECTION
Follow classification and handling rules



4. CYBERSECURITY
Secure devices, networks and software



5. TRANSMISSION DISCIPLINE
Follow authorized radio procedures



6. PHYSICAL SECURITY
Guard personnel, equipment and facilities



7. INCIDENT REPORTING
Report compromise and follow response procedures



8. TRAINING AND COMPLIANCE
Practice procedures and conduct checks



Security Measures and Protocols in Signal Support Operations

Security in signal support operations means protecting communication equipment, networks, and information while maintaining the commander's ability to communicate. It requires both protective measures and procedures that personnel consistently follow.

Communication services remain governed by communication security (COMSEC) policy, regardless of the technology used.

Security Measures and Protocols in Signal Support Operations

1. Follow communication security policies and operating instructions
2. Protect against interception and electronic interference
3. Prepare for compromised equipment and frequencies
4. Use less secure and commercial communication with caution
5. Control access and protect computers and networks
6. Monitor systems and assess vulnerabilities
7. Follow an incident-response plan
8. Coordinate security across the joint force
9. Maintain communication and recovery arrangements

Security Measures and Protocols in Signal Support Operations

1. **Follow communication security policies and operating instructions**

Signal personnel must use the applicable COMSEC policies, standard operating procedures, and communication instructions.

Example: An operator follows the issued communication instructions rather than independently choosing procedures during an operation.

Security Measures and Protocols in Signal Support Operations

2. Protect against interception and electronic interference

The tactical communications are vulnerable to jamming and interception. The signal plan must include protective countermeasures. Electronic protection measures that may include frequency hopping and anti-jamming capabilities. Their employment must follow the applicable plan and instructions.

Why this matter: Communication may be disrupted, or information may be intercepted, even when the equipment remains serviceable.

Security Measures and Protocols in Signal Support Operations

3. Prepare for compromised equipment and frequencies

The communication equipment may fall into an adversary's hands and compromise operational frequencies. It requires planned procedures for changing affected frequencies efficiently.

Example: A lost radio must be treated as a possible security incident. Personnel report it through prescribed channels and implement the directed protective actions.

Lesson: Equipment loss can affect the security of the wider communication network

Security Measures and Protocols in Signal Support Operations

4. Use less secure and commercial communication with caution

The vigilance net operates in a less secure mode and is susceptible to attack and eavesdropping. It identifies ciphertext and double talk as mitigation measures; these should not be interpreted as making the net fully secure

The prudent use of commercial video-conferencing platforms, cellular phones, and satellite phones under existing COMSEC policies.

Lesson: A device or network being available does not mean every type of military information may be transmitted through it

Security Measures and Protocols in Signal Support Operations

5. Control access and protect computers and networks

Security technology	Function
Network Access Control (NAC)	Enforces policies governing access by users and devices.
Unified Threat Management firewall	Combines protections such as filtering, anti-malware, and intrusion detection.
Endpoint security	Protects computers and other endpoints through antivirus, application control, device control, and related measures.
Anti-spam firewall	Provides functions including anti-spam, data-loss prevention, encryption, and threat protection.

Security Measures and Protocols in Signal Support Operations

6. Monitor systems and assess vulnerabilities

The security and monitoring to begin while the network is being built. It describes:

- **Security Operations Center monitoring**: detects and supports reactions to security incidents.
- **Cyber risk assessment**: assesses compliance with the security controls referenced by the manual.
- **Vulnerability assessment and penetration testing**: identifies weaknesses in internal network infrastructure.
- **Security Information and Event Management (SIEM):** collects and analyzes activity across IT resources.

Lesson: Security requires continuing observation and assessment. Installing protective equipment is only one part of the task.

Security Measures and Protocols in Signal Support Operations

7. Follow an incident-response plan

Incident response as action during or after an incident to mitigate its effects and investigate the extent of compromise. CSIRT responsibilities that include maintaining an incident-response plan, investigating incidents, managing communication updates, remediating problems, and recommending improvements.

Example: If suspicious network activity is detected, personnel report it through established procedures. Designated responders investigate and direct the appropriate corrective actions.

Security Measures and Protocols in Signal Support Operations

8. Coordinate security across the joint force

For joint operations, communication interoperability must follow established procedures and protocols. The joint cyberspace operations, incident-response teams from the major services integrate under AFPCyG to coordinate their response. For cyberspace support to AFP joint operations, cyber-unit tasks depend on the Joint Force Commander's mission and intent.

Lesson: Connecting Army, Navy, and Air Force systems also requires coordinated protection and incident-response responsibilities.

Security Measures and Protocols in Signal Support Operations

9. Maintain communication and recovery arrangements

Security includes keeping essential communication available. Redundant links, repair and replacement arrangements, technical support, alternate routing, and primary, alternate, contingency, and emergency plans

Secure signal support combines approved procedures, protected equipment and networks, monitoring, coordinated incident response, and recovery arrangements so the force can continue its mission despite disruption or compromise.

End of Presentation